GGG-A-926c
JULY 13, 1961
SUPERSEDING
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August 9, 1956

FEDERAL SPECIFICATION

AX, (SINGLE BIT, DOUBLE BIT, PICK HEAD AND MATTOCK HEAD)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers various types of axes for chopping and splitting wood, clearing land of under brush, fire fighting and a utility tool for use on lifeboats and liferafts (see 6.1).

1.1.1 *Federal specification coverage.* Federal specifications do not include all types and classes of the commodity indicated by the titles of the specifications, but cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types, classes, designs, and styles.* Axes shall be of the following types, classes, designs, and styles, as specified (see 6.2):

Type I.—Chopping.
- Class 1.—Single bit.
  - Design A.—Men's (Dayton pattern).
  - Design B.—Boys'.
  - Design C.—House.
  - Design D.—Hunter's.
  - Design E.—Belt.
    - Style 1.—Wood handle.
    - Style 2.—Steel handle.
- Class 2.—Double bit.
  - Design A.—Men's (Western pattern).
  - Design B.—Cruiser (Crown or Michigan pattern).
  - Design C.—Swamping.

Type II.—Pick head (fire or fireman's).

Type III.—Mattock head (Pulaski).

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:

NN-H-93—Handles; Hickory, Striking-Tool.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL–P–15424 — Packaging of Hand Tools for Domestic and Overseas Shipment and Storage.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Ax head.* The head shall be forged in one piece of a suitable alloy or carbon steel with not more than 0.05 percent sulphur nor 0.04 percent phosphorus content.

3.1.2 *Handles.* Except as specified in 3.7.1.6.2, handles shall be of hickory wood, grade AA or A conforming to the material, workmanship, and general requirements of Federal Specification NN–H–93.

3.2 Construction.

3.2.1 *Blades or bits.* The blades or bits shall have a conventional taper from the eye section terminating in a ground-roll-beveled cutting edge, sharpened, and ready for use. The blades or bits shall be properly hardened and tempered throughout and shall withstand the test specified in 4.4.2. Except as specified for type I, class 1, design E, style 2 (see 3.7.1.6.2), blades or bits shall be hardened to not less than 53 nor more than 58 on the Rockwell "C" scale for a distance of not less than 1 inch back from the cutting edge (see 4.4.1).

3.2.2 *Eye.* The eye of all axes except type I, class 1, design E, style 2 (see 3.7.1.6.2) shall be of such size and strength as to prevent buckling, cracking, breaking, or spreading in ordinary use. The eye shall be elliptical in shape and tapered from the larger opening at the top or wedge end to a smaller opening at the handle insertion end to provide for securing the head to a handle with wedge or wedges. The edges of the eye shall be free of all burrs to provide for proper insertion of the handle. The difference in the thickness of corresponding sections of opposite walls of the eye shall not exceed $\frac{1}{32}$ inch. The median line of the eye shall be in proper alignment with the cutting edge of the ax. The eye shall be hardened to not more than 30 on the Rockwell "C" scale (see 4.4.1).

3.2.3 *Assembled hickory handles.* Hickory handles shall be properly fitted and secured to the head with a chemical adhesive bond, wood wedges, or wood wedges in combination with one or more horizontally or diagonally corrugated metal wedges or metal screws unless a specific method of assembly is specified (see 6.2). Wedges and screws shall be driven or screwed, respectively, flush within plus or minus $\frac{1}{32}$ inch with the front surface of the eye. The wedge, wedges, or screws, or a combination of the same, shall create sufficient expansion to fill the eye and tightly wedge the handle in the eye of the ax, so as to minimize the head from working loose under the most severe working conditions. The screws shall be of the tapered headless type. Handles shall be smoothly sanded and show no evidence of sander streaks or tool marks. Unless otherwise specified (see 6.2), handles may be flame treated or color stained and, at the option of the manufacturer, shall be coated with either a transparent lacquer or

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wax that does not conceal the annual rings of growth and grain of the wood.

3.2.3.1 *Assembled hickory handles for United States Forest Service.* When specified (see 6.2), handles shall be assembled for the United States Forest Service as follows:

Handles shall be properly fitted and secured with soft-wood wedges only. The end of the handle may be split or sawed with a thin blade for wedging. The width of the sawed slot shall not exceed 7/64 inch. Both the handle and the wedge shall have a moisture content of not more than 8 percent at the time of fitting and assembling the handle to the ax. The excess handle and wedges shall be trimmed flush within plus or minus 1/32 inch with the front surface of the eye. Handles shall be smoothly sanded and shall show no evidence of sander streaks or tool marks. Handles shall be furnished free of lacquer, paint, varnish, shellac, waxes, or other coatings.

3.2.4 *Chemical adhesive bond.* Handles secured by chemical-adhesive bond shall be securely bonded to the head by means of a chemical adhesive that will prevent the handle from loosening under any normal working conditions. Axes having such chemical adhesive shall be subjected to the pull apart test specified in 4.4.6.

3.2.5 *Illustrations.* The illustrations shown herein are descriptive and are not restrictive, and are not intended to preclude the purchase of axes otherwise conforming to this specification.

3.3 Finish.

3.3.1 *Surface finish.* The bit end of all axes except type I, class 1, design E, style 2 (see 3.7.1.6.2) shall have a polished finish for not less than 3/8 inch back from the cutting edge. The finish on all polished surfaces shall be not coarser than that resulting from the use of an 80-grit abrasive belt or have a maximum surface roughness of 50 microinches. The sides of the blade or bit, and body of all axes, except polished surfaces, shall have a smoothly ground finish not coarser than that resulting from the use of a 60-grit abrasive belt or have a maximum surface roughness of 57 microinches. Front and back edges of axes shall have all flash removed and a finish not coarser than that resulting from the use of a 30-grit abrasive belt or have a maximum surface roughness of 125 microinches. A specimen block of equal material and hardness as that of the ax and of suitable size may be used for comparison to determine compliance. One side of the specimen block shall be finished by using a new abrasive belt of the proper grit, running dry. By use of the specimen block the finished area of the ax need not be checked to determine the roughness height rating in microinches. Determination for compliance of finishes specified shall be made on the finished product.

3.4 Coatings. Blades or bits, and bodies of all axes, except polished surfaces, shall be neatly coated with a waterproof paint, lacquer, or enamel. The coating shall not be tacky nor flake off readily. Polished surfaces shall be coated with a clear lacquer or other suitable transparent coating applied to either the metal or entire surface to deter rusting in transit or in short term storage. In application the clear lacquer coating may extend over other surfaces of the ax but shall not be substituted for another coating.

3.5 Weight. The weight specified for all types of axes except type I, class 1, design E, style 2 (see table II) shall be exclusive of the handle.

3.6 Identification marking. Each ax head shall be permanently marked in a legible manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined. Axes of type I, class 1, design A, men's single bit and type I, class 2, men's double bit patterns shall also be permanently marked with the weight of head.

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3.7 Type I, chopping.

3.7.1 *Class 1, single bit.* The head of single-bit axes shall have a cutting bit or blade on one end and a flat or slightly convexed rectangular surface defined as the "top face" on the opposite end. The top face shall have a polished surface and, except for design E, style 2, shall be hardened to not more than 30 on the Rockwell "C" scale when tested as specified in 4.4.1. Unless otherwise specified (see 6.2), all single bit axes except design E, style 2 (see 3.7.1.6.2) shall be furnished fitted with curved wooden handles.

3.7.1.1 *Design A, mens (Dayton pattern).* The ax shall be of the design known commercially as "Dayton". The ax shall be similar to figure 1.

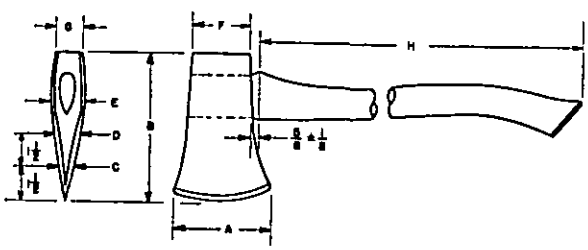


FIGURE 1.—Type I, class 1, designs A, B and C, single bit, chopping ax, men's (Dayton pattern), boys and house.

3.7.1.2 *Design B, boys'.* Design B ax shall be of the design known commercially as "boys", and shall be similar to figure 1.

3.7.1.3 *Design C, house.* Design C ax shall be of the design known commercially as "house", and shall be similar to figure 1.

3.7.1.4 Type I, class 1, designs A, B, and C axes shall conform to table I, as applicable, for the weight of head specified (see 6.2).

3.7.1.5 *Design D, hunter's.* Design D ax shall be of the design known commercially as "hunter's", and shall be similar to figure 2.

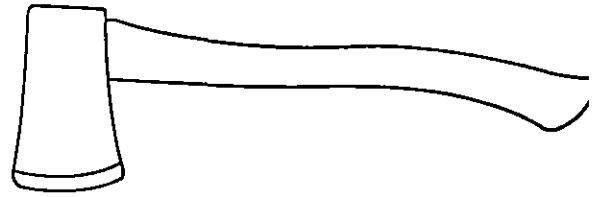


FIGURE 2.—Type I, class 1, design D, hunter's, single bit, chopping ax.

3.7.1.5.1 *Sheaths.* When specified (see 6.2), axes shall be fitted with sheaths which, unless otherwise specified (see 6.2), shall be of leather. Sheaths shall be of substantial construction in order to withstand long, hard use. They shall have a full-size flap fitted with a strong brass or corrosion-resistant steel snap fastener and at least two belt loops, or two formed slits approximately 1⅝ inches long, through which to pass a carrying belt. If canvas sheaths are specified (see 6.2), the canvas shall be mildewproofed.

3.7.1.6 *Design E, belt.* The ax shall be of the design known commercially as "belt". At the option of the manufacturer, unless otherwise specified (see 6.2), the head may have a beveled slot on the handle side of the head approximately midway between the eye and cutting edge, to provide for the drawing of nails. When specified (see 6.2), axes shall be fitted with sheaths in conformance with 3.7.1.5.1. Unless otherwise specified (see 6.2), the handle shall be furnished either with or without a hole for a lanyard or thong.

3.7.1.6.1 *Style 1, wood handle.* Style 1 belt ax shall be similar to figure 3.

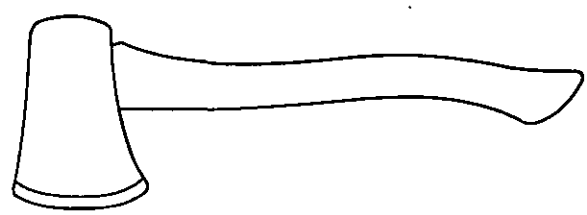


FIGURE 3.—Type I, class 1, design E, style 1, wood handle, belt, single bit, chopping ax.

3.7.1.6.2 *Style 2, steel handles.* The head of the ax shall be hardened to not less than 50

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TABLE I.—*Type I, class 1, chopping, single bit, designs A, B and C, men's (Dayton or Michigan pattern), boys', and house*

Design	Head								Handle H
	Weight	Width at cutting edge A	Overall length B	Thickness C	Thickness D	Thickness at widest point E	At top face		
							Width F	Thickness G	
	±3 ounces	±1/4 inch	Maximum	±1/16 inch except as noted	±1/16 inch except as noted	±1/16 inch except as noted	±1/4 inch	±1/16 inch	±1/2 inch
	Pounds	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*
A	3 1/2 4 5	4 11/16 4 3/4 5	7 1/2 7 3/4 8 1/4	3/8 7/16 1/2	11/32 5/8 9/16	1 1/8 1 3/16 1 1/4	3 1/8 3 1/4 3 1/4	1 3/32 1 1/16 1 1/8	31 31 31
B	2 1/4	4	6 1/2	5/16±1/8	23/32±1/8	1 1/32±1/8	2 11/16	1	24
C	2 1/4	4	6 1/2	3/8±1/8	23/32±1/8	1 1/32±1/8	2 11/16	1	15 1/8

nor more than 58 on the Rockwell "C" scale for a distance of not less than 3/4 inch back from the cutting edge of the blade and not less than 45 nor more than 53 on the top face when tested as specified in 4.4.1. The blade or bit end shall have a polished surface for not less than 3/16 inch back from the cutting edge. The axes shall have curved steel handles which shall be a forged or welded extension of the head or shall be seamless or welded steel tubing securely fastened to the head by pins, swaging, silver braze fuzing, chemical adhesive or by internal expansion in heads with conventional eyes. There shall be no openings between the handle and head where the eye for the wood handle is normally located. At least 6 inches of the length from the gripping end shall be built up of flat individual leather washers or shall be covered with rubber, fiber, neoprene, vinyl, vinyl-nylon or equally comparable material. The leather washers shall be flat, elliptical shaped, and of a good grade of sole leather or similar material. They shall be not less than 1/8 inch thick. The grip material shall be permanently and securely attached to the metal handle and shall be smoothly finished and conform generally to the shape as the same type specified in Federal Specification NN–H–93 for wood handles. Tubular steel handles shall be subjected to the pull apart test specified in 4.4.6. Steel handles shall be subjected to the bending test in 4.4.5.

3.7.1.7 Type I, class 1, designs D and E, styles 1 and 2 axes shall conform to the applicable requirements of table II for the weight of head specified (see 6.2). Style 2 ax shall be similar to figure 4 or 5 except that the striking face of the head shall be either

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as shown on figures 4 or 5 or convex as shown on figure 3.

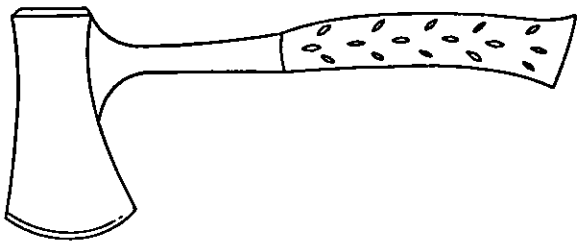


FIGURE 4.—Type I, class 1, design E, style 2, steel handle, belt, single bit, chopping ax.

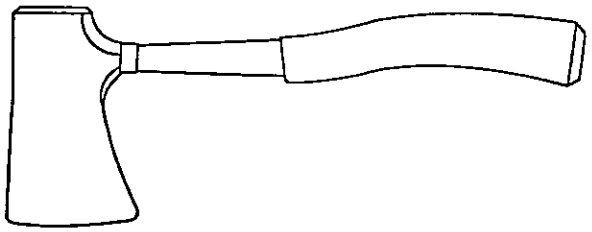


FIGURE 5.—Type I, class 1, design E, style 2, steel handle, belt, single bit, chopping ax.

3.7.2 *Class 2, double bit.* The head of double-bit axes shall have identical cutting blades or bits on both ends. The distance from the center line of the eye to each cutting edge shall not differ more than ⅛ inch to insure proper balance. All double-bit axes shall be furnished fitted with straight handles.

3.7.2.1 *Design A, men's (Western pattern).* Design A, men's (Western pattern) ax shall be of the design known commercially as "Western". The ax shall be similar to figure 6 and shall conform to table III for the weight of head specified (see 6.2).

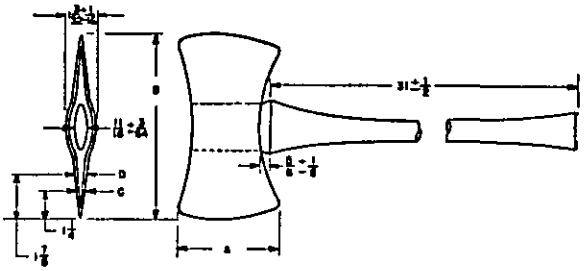


FIGURE 6.—Type I, class 2, design A, men's (Western pattern), double bit, chopping ax.

3.7.2.2 *Design B, cruiser (Crown or Michigan pattern).* Design B, cruiser (Crown or Michigan pattern) ax shall be of the design known commercially as "Crown" or "Michigan". The ax shall conform to figure 7.

TABLE II.—*Type I, class 1, chopping, single bit, design D, hunter's; and design E belt, styles 1 and 2, wood and steel handles*

Design and style	Head					Overall length[1] of handle with head assembled
	Weight	Width at cutting edge	Length overall	At top face		
				Width	Thickness	
	±3 ounces	±3/16 inch	±¼ inch	±¼ inch	±3/16 inch	±⅝ inch
	Pounds	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*
D	1¼	3⅜	5⅛	2⅛	⅞	14
	1¾	3⅝	5¾	2⅜	⅞	16½
E, style 1	1¼	3¼	5⅛	2	⅞	13⅝
E, style 2	[2]1¾	3	4⅞	1⅞	13/16	13⅛

[1] Handle length shall include the portion inserted in the head.

[2] Weight includes handle.

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TABLE III.—*Type I, class 2, chopping, double bit, design A, men's (Western pattern)*

Weight ±3 ounces	Head				
	Width at cutting edge A ±¼ inch	Overall length[1] B Maximum	Thickness C ±1/16 inch	Thickness D ±⅛ inch	Thickness at widest point E ±1/16 inch
Pounds	*Inches*	*Inches*	*Inch*	*Inch*	*Inches*
3½	4¾	10⅝	⅝	½	1
4	5	10¾	⅝	½	1 3/32

[1] From center of eye to each cutting edge shall be equal within 1/16 inch to insure proper balance.

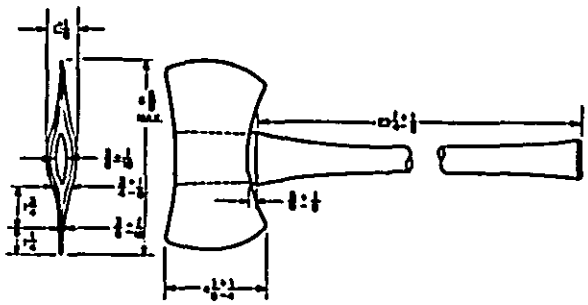


FIGURE 7.—Type I, class 2, design B, cruiser (Crown or Michigan pattern), double bit, chopping ax.

3.7.2.3 *Design C, swamping.* Design C, swamping ax shall be of the design known commercially as swamping. The ax shall conform to figure 8.

3.8 **Type II, pick head (fire or fireman's).** The pick-head ax known commercially as "fire or fireman's" shall have a head with an ax blade or bit on one end and a pick on the other end. The ax blade or bit shall conform to the requirements specified in 3.2.1. The pick end shall be hardened to not less than 53 nor more than 58 on the C scale at approximately ½ inch from the point when tested as specified in 4.4.1. The pick end shall have a polished surface for not less than ½ inch back from the point. The ax shall be furnished fitted with a curved handle. The handle and unpolished surfaces of the head shall be coated with a durable red paint or red enamel. The ax shall conform to table IV and shall be similar to figure 9.

3.9 **Type III, mattock head (Pulaski).** Mattock head ax (known commercially as "Pulaski") shall have a head with an ax blade or bit on one end and a grubbing blade set at right angles to the perpendicular axis of the eye on the other end and shall be fitted with a straight double-bitted ax handle. The ax blade or bit shall conform to the requirements specified in 3.2.1. The grubbing blade shall terminate in a single bevel of not less than 5/16-inch width, and have a roll-ground cutting edge sharpened and ready for service. The unpolished surfaces of the concaved side of the grubbing blade shall have a ground finish not coarser than that resulting from the use of an abrasive wheel of not less than 60 grit for a distance of not less than 4 inches back from the cutting edge. The entire convexed side shall have the same grade of finish. The top flat portion of the eye section and the fillet between the eye and concaved side of the grubbing blade shall have forging flash removed. The grubbing blade shall be hardened to not less than 50 nor more than 55 on the Rockwell "C" scale for a distance of not less than 2 inches back from the cutting edge when tested as specified in 4.4.1. The ax shall conform to figure 10.

3.10 **Workmanship.** Workmanship shall be of the highest grade and shall be equal in

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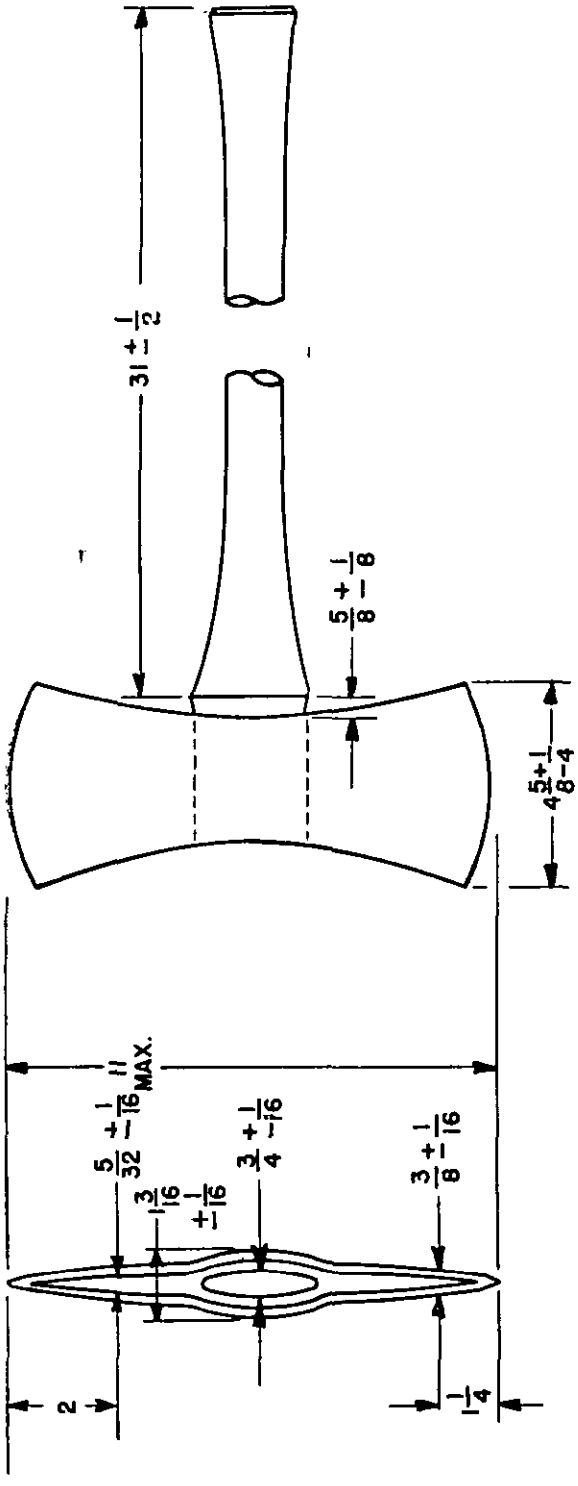


FIGURE 8.—Type II, class 2, design C, swamping, double bit, chopping ax.

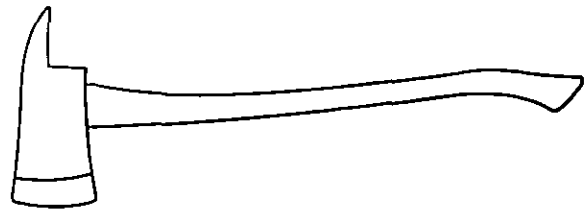


FIGURE 9.—Type II, pick-head (fire or fireman's) ax.

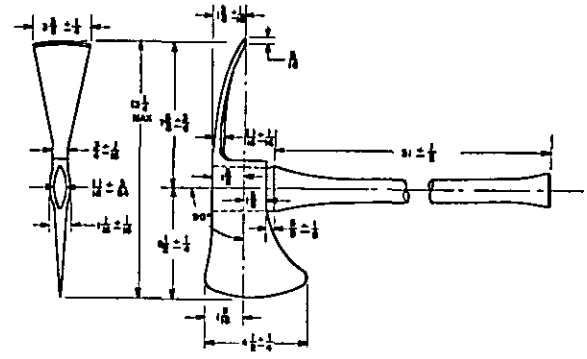


FIGURE 10.—Type III, mattock head (Pulaski) ax.

every respect to good commercial practice. Axes shall be free from laps, cracks, pipes, seams, or other defects which may impair *their serviceability or durability. Small scale* pits of an average diameter of less than 1/8 inch shall be permissible in the concaved areas adjacent to the eye.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of *the examination and tests shall be kept com*plete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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TABLE IV.—*Type II, pick head (fire or fireman's) ax*

Weight	Head				Overall length[1] of handle with head assembled
	Width of ax blade cutting edge	Length			
		Overall	From center of eye to cutting edge	From center of eye to end of pick	
±4 ounces	±1/8 inch	±1/8 inch	±7/16 inch	±7/16 inch	±1/8 inch
Pounds	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*
6	5	12	6	6	34 1/2

[1] Handle length shall include portion inserted in the head.

4.2 Sampling for acceptance inspection.

4.2.1 *Inspection lot.* All axes of the same type, class, design, style, and weight offered for delivery at one time shall be considered a lot for purposes of sampling and tests.

4.2.2 *Sampling for visual and dimensional examination.* A random sample of axes shall be selected from each inspection lot of material offered for examination of visual and dimensional characteristics with lot acceptance based on the sampling inspection requirements in accordance with table V and Military Standard MIL–STD–105.

TABLE V.—*Sampling for visual and dimensional examination AQL (approx.) = 1.5 percent defective*

Number of axes in inspection lot	Number of axes in sample
40 and under	Inspection level III
41 to 800	Inspection level II
801 and over	Inspection level I

Note.—The inspector may institute tightened or reduced inspection in accordance with section 9 of Military Standard MIL–STD–105.

4.2.3 *Sampling for tests other than accelerated cutting test.* A random sample of axes shall be selected from each inspection lot in accordance with Military Standard MIL–STD–105 at inspection level L–8 and shall be subjected to the applicable tests specified in 4.4.1, 4.4.2, 4.4.3, 4.4.5 and 4.4.6. The Acceptable Quality Level shall be equal to 4.0 percent defective.

4.2.4 *Sampling for accelerated cutting tests.* A random sample of axes except type I, class 1, design E, style 2 ax shall be selected from each inspection lot in accordance with Military Standard MIL–STD–105 at Inspection Level L–4 and shall be subjected to the test specified in 4.4.4. If any sample ax fails the test, the lot shall be rejected.

4.3 Examination. Each of the sample axes selected in accordance with 4.2.2 shall be examined to verify compliance with this specification. Examination shall be made to determine if the handles have been properly and tightly fitted in the heads. Examination shall be conducted as specified in table VI. Any ax in the sample that fails to pass the examination of 4.3 or the tests of 4.4 shall be rejected, and if the number of defective axes in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

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TABLE VI.—*Classification of defects in accordance with Military Standard MIL–STD–105*

Categories	Defects
Critical:	
1	Handle not properly fitted in head to prevent ax head from working loose or flying off handle in service; wedges missing, damaged, nonconforming or improperly installed in end of handle.
Major:	
101	Type, class, design and style not as specified.
102	Ax head material not as specified, not free from laps, cracks, pipes, seams or excessive scale pits; welding, when applicable, nonconforming or evidence of incomplete fusion.
103	Gripping surface not smooth, material not as specified, not securely fastened to metal, length less than allowable minimum (type I, class 1, design E, style 2 only).
104	Metal handle nonconforming, not curved as required (type I, class 1, design E, style 2 only).
105	Hand grip surface not free from coating of any nature, (type I, class 1, design E, style 2 only).
106	Handle not smoothly finished, evidence of splinters, sander streaks or toolmarks, not of the specified grade of hickory; shape nonconforming, not curved or straight as applicable.
107	Handle not free from lacquer, paint, varnish, shellac, waxes or other coatings (except type II).
108	Handle not coated as required, paint not smooth or adherent, color of paint not as specified (type II only).
109	Shape of ax head, cutting blade(s) and top face as applicable not as specified.
110	Median line of eye not in proper alignment with cutting edge (except type I, class 1, design E, style 2).
111	Wedge and handle not assembled flush with top of eye, (except type I, class 1, design E, style 2).
112	Cutting edge nonconforming; not ground as required, not sharp, evidence of nicks, burrs or grinding burns.
113	Pick end or grubbing blade nonconforming, damaged, not finished or sharpened as required (type II and III only).
114	Beveled slot for drawing of nails missing, nonconforming, or not located as specified (type I, class 1, design E only).
115	Eye shape and taper not as specified: eye section not of sufficient proportion to prevent buckling, cracking, breaking, or spreading during service, (except type I, class 1, design E, style 2).
116	Sheath when specified, missing, damaged, nonconforming or not of the required material (type I, class 1, design D, and design E, style 1 only).
Minor:	
201	Difference in thickness of corresponding sections of opposite walls of the eye exceeds the allowable limit, (except type I, class 1, design E, style 2).

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TABLE VI.—*Classification of defects in aacordance with Military Standard MIL–STD–105 (continued)*

Categories	Defects
Minor:	
202	Width of cutting edge not within the specified tolerance.
203	Width of head at top face not within the required tolerance, (type I only).
204	Thickness of head at top face not within the specified limits (type I only).
205	Overall length of head not within allowable limits.
206	Length from center of eye to ax blade not within the specified tolerances (types II and III only).
207	Length from center of eye to end of pick or grubbing blade as applicable, not within the required limits (types II and III only).
208	Overall length of handle including head not within the specified tolerance.
209	Cutting edge area as specified not polished, or polished surface finish coarser than allowable.
210	Unpolished metal surfaces not coated as specified to prevent corrosion; coating tacky or nonadherent.
211	Marking, weight and manufacturer's identification missing, incorrect, illegible or not permanent.
212	Cleaning, preservation, packaging not as required.

4.4 Test procedures.

4.4.1 *Hardness.* Sufficient grinding shall be done on the areas to be tested to insure that all carburization is removed and that the areas are parallel in order to insure a proper reading. At least two hardness readings shall be taken on the specified portion or portions of each sample ax. The values thus obtained shall comply with the requirements of 3.2.1, 3.2.2, 3.7.1, 3.7.1.6.2, 3.8, and 3.9.

4.4.2 *Mallet test.* The ax head shall be placed flat on a bench or other flat support so that the bit end from the lower portion of the eye overhangs the edge of the bench or support. The remainder of the head on the bench shall be manually or mechanically held. Two hard blows shall be struck with a 3½- or 4-pound wooden mallet on the portion of the head which overhangs the edge of the bench or support. The test shall be repeated on the opposite side. There shall be no fracture or deformation of the metal as a result of the test (see 3.2.1).

4.4.3 *Practical cutting test.* Ax blades shall cut easily through hardwood knots. After striking a minimum of ten heavy blows with each sample ax on hardwood there shall be no evidence of chipping, dulling, or turning over of the cutting edges, loosening of handles or wedges, or any other damage to the ax that indicates inferior material or workmanship.

4.4.4 *Accelerated cutting test.* Each sample ax, selected in accordance with 4.2.4, shall be prepared by grinding the blade or blades to a double-beveled chisel edge having an included angle of from 60 to 80 degrees for the purpose of this test. Care shall be taken dur-

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ing the grinding operation to prevent drawing the temper of the steel, and the test edge shall be ground sharp and true. The cutting edge or edges of each sample ax shall be subjected to five hard blows on a 16-penny (8-gage 0.162 inch) common steel nail(s), or equivalent, resting on a block of hardwood. Failure shall have occurred should the cutting edge show evidence of spalling, cracking, chipping, turning, or flattening.

4.4.5 *Steel handle bending test.* Samples of type I, class 1, design E, style 2 axes shall show no permanent deformation of the handle in excess of ⅛ inch when the head is securely clamped and a load of 250 pounds is applied 11 inches from the center of the head in a parallel direction to the axis of the ax as shown in figure 11.

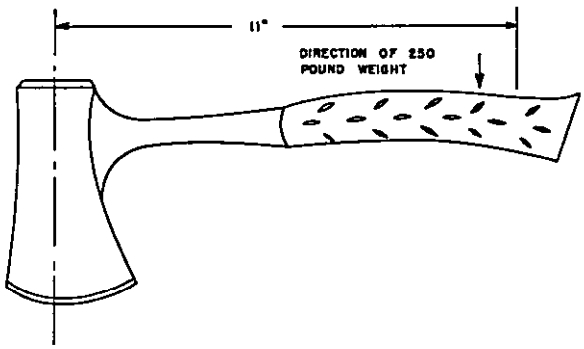


FIGURE 11.—Direction of weight applied to type I, class 1, design E, style 2, ax.

4.4.6 *Pull apart test.* Axes with tubular steel handles or hickory handles assembled with chemical adhesive bond shall be subjected to this test. Each of the sample axes having tubular steel handles shall have the grip removed and a steel plug inserted in the end of the tube to support the tube when gripped. The head and end of the handle shall be securely held in a standard tensile testing machine or some other method shall be employed. The ax shall not pull apart or loosen when subjected to a pulling pressure of 3,500 pounds for axes with head weight 2½ pounds and over nominal weight and 2250 for axes with head weight less than 2½ pounds nominal weight.

4.5 **Possible test failures.** Possible test failures are defined as follows:

(*a*) *Weight.*
- (1) Weight not within specified tolerance.

(*b*) *Hardness.*
- (1) Hardness of cutting edge, and pick or grubbing blade when applicable, not within specified values.
- (2) Hardness of eye section exceeds the allowable maximum value.
- (3) Hardness of top face not within the specified values (type I, class 1, design E, style 2).

(*c*) *Mallet.*
- (1) Evidence of fracture or deformation of ax head when subjected to the specified test.

(*d*) *Practical cutting test.*
- (1) Evidence of chipping, dulling or turning over of cutting edge, loosening of handle or wedges or other damage when subjected to the specified test.

(*e*) *Accelerated cutting test.*
- (1) Evidence of spalling, cracking, chipping, turning or flattening of the cutting edge when ground and tested as specified.

(*f*) *Steel handle bending test.*
- (1) Evidence of permanent deformation of the handle when head is clamped and the specified load is applied as required (type I, class 1, design E, style 2 only).

5. PREPARATION FOR DELIVERY

(Civil agencies. Federal Standard No. 102 should be referred to for definitions of the various levels of packaging protection for supplies and equipment.)

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5.1 Cleaning, preservation and packaging.

5.1.1 *Level A.* Axes shall be cleaned, preserved and packaged in accordance with level A in Military Specification MIL–P–15424.

5.1.2 *Level C.* Cleaning, preservation and packaging shall be in accordance with the contractor's commercial practice.

5.2 Packing.

5.2.1 *Level A.* Axes, cleaned, preserved and packaged by level A or C as specified (see 6.2), shall be packed in accordance with the level A requirements of Military Specification MIL–P–15424.

5.2.2 *Level B.* Axes, cleaned, preserved and packaged by level A or C as specified (see 6.2), shall be packed in accordance with the level B requirements of Military Specification MIL–P–15424.

5.2.3 *Level C.* Axes, cleaned, preserved and packaged as specified (see 6.2), shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with rules or regulations of other carriers as applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Military marking.* In addition to any special marking required by the contract or order herein, interior and exterior shipping containers shall be marked in accordance with Military Specification MIL–P–15424.

5.3.2 *Civil agencies.* Marking for shipment shall be in accordance with the standard marking of Federal Standard No. 123.

6. NOTES

6.1 Intended use.

6.1.1 All type I axes, except class 1, design E, style 2 are intended for general utility use in forestry work.

6.1.2 Type I, class 1, design E, style 2 axes are intended for use as a utility tool on lifeboats and life rafts.

6.1.3 Type II axes are intended for use by firefighters.

6.1.4 Type III axes are intended for use in clearing grounds of underbrush.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type, class, design, and style of ax required (see 1.2).
(*c*) Handle coating or finish other than specified (see 3.2.3).
(*d*) Specify if handles are to be assembled by a certain method (see 3.2.3).
(*e*) Specify if handles are to be assembled for United States Forest Service requirements (see 3.2.3.1).
(*f*) Weight of head required (see 3.7.1.4, 3.7.1.7 and 3.7.2.3).
(*g*) Specify if straight handles are required for any class 1 axes (see 3.7.1).
(*h*) Specify whether sheaths are required and if material other than leather is required (see 3.7.1.5.1 and 3.7.1.6).
(*i*) Specify if heads are to be furnished with a beveled slot (see 3.7.1.6).
(*j*) If a hole for a lanyard is required for type I, design E belt ax, state the size and location (see 3.7.1.6).
(*k*) Level of cleaning, preservation, packaging, and packing required (see section 5).
(*l*) Marking, if different (see 5.3).

6.3 This specification conforms to the patterns, weights, and tolerances of the National Bureau of Standards Publication Simplified

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Practice Recommendation R158–42 for Axes, Forged.

6.4 **Transportation description.** The applicable transportation description is:

Axes, with handles:
Carload minimum weight 30,000 pounds.

Axes, handles:
Truckload minimum weight 30,000 pounds.

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MILITARY CUSTODIANS:
Army—Quartermaster
Navy—Bureau of Ships
Air Force—Mobile Air Materiel Area.

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